

Debconf Full Guide for Package Automation

debconf is a configuration management system used by Debian and Ubuntu packages to ask configuration questions during package installation.

Core Concepts

- **debconf database:** Stores configuration selections for packages.
 - **Preseed files / selections:** Automate the installation by feeding answers.
 - **Templates:** Define the questions a package can ask (priority, type, description, etc.).
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View Debconf Database

Get Current Setting

```
bash sudo debconf-show <package-name>
```

Example:

```
bash sudo debconf-show grub-pc
```

Dump All Settings

```
bash sudo debconf-get-selections
```

To dump only non-default settings:

```
bash sudo debconf-get-selections | grep -v '^s*$' > mydebconf-selections.txt
```

Tip: Useful for backing up or replicating configurations.



Set/Preconfigure a Debconf Value

You can programmatically set values using `debconf-set-selections`.



Syntax:

```
bash echo "<package> <question> <type> <value>" | sudo debconf-set-selections
```

Example:

```
bash echo "grub-pc grub-pc/install_devices multiselect /dev/sda"  
| sudo debconf-set-selections
```

```
Type: boolean - string - select - multiselect - note - password
```

Find Specific Questions/Settings

Option 1: Extract Debconf Templates from .deb File

```
bash dpkg-deb -e <package>.deb tempdir/ cat tempdir/templates
```

Option 2: Use apt-get in Debug Mode

```
bash sudo apt-get install -y <package> -o Dpkg::Options::="--  
force-confask"
```

Option 3: Query Debconf Database

```
bash sudo debconf-get-selections | grep <package-name>
```

Export and Reapply Settings

Step 1: Export

```
bash sudo debconf-get-selections > selections.txt
```

Step 2: Reapply on Another Machine

```
bash sudo debconf-set-selections < selections.txt
```

Non-Interactive Installations

```
bash export DEBIAN_FRONTEND=noninteractive sudo apt-get install -  
y <package>
```

Testing Debconf Interaction of a Package

Text-based:

```
bash sudo dpkg-reconfigure -freadline <package>
```

GUI:

```
bash sudo dpkg-reconfigure <package>
```

Reset/Delete Debconf Settings

Interactive:

```
bash sudo debconf-communicate
```

Then type:

```
PURGE <package-name>
```

CLI:

```
bash echo "PURGE grub-pc" | sudo debconf-communicate
```

Use in Scripts

```
bash echo "postfix postfix/mailname string example.com" | sudo
debconf-set-selections echo "postfix postfix/main_mailer_type
string 'Internet Site'" | sudo debconf-set-selections
DEBIAN_FRONTEND=noninteractive sudo apt-get install -y postfix
```

Notes and Tools

- debconf-get-selections is in debconf-utils:

```
bash sudo apt install debconf-utils
```
 - debconf-set-selections is part of debconf.
 - Templates after install: `/var/lib/dpkg/info/<package>.templates`
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Useful Files & Paths

Path	Description
<code>/var/cache/debconf/config.dat</code>	The main debconf database
<code>/var/lib/dpkg/info/<pkg>.templates</code>	Template questions (once installed)
<code>/etc/debconf.conf</code>	Debconf backend config

Summary Cheat Sheet

```
```bash
```

# Show current package questions

```
sudo debconf-show
```

# Dump all selections

```
sudo debconf-get-selections
```

# Set a configuration answer

```
echo " " | sudo debconf-set-selections
```

# Export selections

```
sudo debconf-get-selections > selections.txt
```

# Import selections

```
sudo debconf-set-selections < selections.txt
```

# Reconfigure package

```
sudo dpkg-reconfigure
```

# Install without prompts

```
DEBIAN_FRONTEND=noninteractive apt install -y ``
```